

CS2601 Linear and Convex Optimization

Homework 4 Solution

Due: 2021.10.28

1. Let Δ_{n-1} be the probability simplex. The **entropy** of a probability distribution $\mathbf{x} \in \Delta_{n-1}$ is defined by

$$H(\mathbf{x}) = - \sum_{i=1}^n x_i \log x_i.$$

where we use the convention $0 \log 0 = 0$.

- (a). Use the concavity of $\log x$ to show that $H(\mathbf{x}) \leq \log \|\mathbf{x}\|_0 \leq \log n$, where $\|\mathbf{x}\|_0 = \sum_{i=1}^n \mathbb{1}\{x_i \neq 0\}$ is the number of nonzero components of $\mathbf{x}$. Hint: If $\|\mathbf{x}\|_0 = k$, you can assume the first k components are nonzero without loss of generality.
- (b). Show that the uniform distribution $\bar{\mathbf{x}}$ with $\bar{x}_i = \frac{1}{n}$ for $i = 1, 2, \dots, n$ is the **unique** maximum of $H(\mathbf{x})$ on Δ_{n-1} . Hint: For uniqueness, show $H(\mathbf{x})$ is strictly concave on $C = \{\mathbf{x} \in \Delta_{n-1} : \mathbf{x} > \mathbf{0}\}$ and use part (a) for $\mathbf{x} \in \Delta_{n-1} \setminus C$.

Proof. (a). Note that $\log x$ is concave. Suppose $\|\mathbf{x}\|_0 = k \leq n$ and the first k components are positive. Then

$$H(\mathbf{x}) = \sum_{i=1}^n x_i \log \frac{1}{x_i} = \sum_{i=1}^k x_i \log \frac{1}{x_i} \leq \log \left(\sum_{i=1}^k x_i \frac{1}{x_i} \right) = \log k = \log \|\mathbf{x}\|_0 \leq \log n.$$

(b). Note

$$H(\bar{\mathbf{x}}) = - \sum_{i=1}^n \frac{1}{n} \log \frac{1}{n} = \log n \geq H(\mathbf{x}), \quad \forall \mathbf{x} \in \Delta_{n-1}$$

so $\bar{\mathbf{x}}$ is a maximum of H on Δ_{n-1} . For $\mathbf{x} \in C$,

$$-\nabla^2 H(\mathbf{x}) = \text{diag} \left\{ \frac{1}{x_1}, \dots, \frac{1}{x_n} \right\} \succ \mathbf{0}$$

so H is strictly concave on C (you can also use the remark in problem 4 of HW3), and $\bar{\mathbf{x}}$ is the unique maximum of H on C . For $\mathbf{x} \in \Delta_{n-1} \setminus C$, since $\|\mathbf{x}\|_0 \leq n-1$,

$$H(\bar{\mathbf{x}}) = \log n > \log \|\mathbf{x}\|_0 \geq H(\mathbf{x}),$$

so it is not a maximum. Therefore, $\bar{\mathbf{x}}$ is the unique maximum of H on Δ_{n-1} .

□

2. Let $f : (a, b) \rightarrow \mathbb{R}$ be convex, where $-\infty \leq a < b \leq +\infty$. Let X be random variable taking values in (a, b) . Suppose the expectations $\mathbb{E}X$ and $\mathbb{E}f(X)$ exist. Prove Jensen's inequality $f(\mathbb{E}X) \leq \mathbb{E}f(X)$ by completing the following steps.

(a). Let $\mu = \mathbb{E}X$. For $a < s < \mu < u < b$, show

$$\frac{f(\mu) - f(s)}{\mu - s} \leq \frac{f(u) - f(\mu)}{u - \mu}.$$

(b). Show that there exists $\beta \in \mathbb{R}$ such that

$$f(x) \geq f(\mu) + \beta(x - \mu), \quad \forall x \in (a, b) \quad (\star)$$

Hint: You can take

$$\beta = \sup_{a < s < \mu} \frac{f(\mu) - f(s)}{\mu - s}.$$

Obviously $\beta > -\infty$. Use part (a) to show that $\beta < +\infty$ and satisfies $(\star)$ (consider $a < x < \mu$ and $\mu < x < b$ separately).

(c). Show that

$$f(X) \geq f(\mu) + \beta(X - \mu).$$

and conclude $\mathbb{E}f(X) \geq f(\mathbb{E}X)$ by taking expectation.

Remark. If f is differentiable, we can take $\beta = f'(\mu)$ by the first-order condition. Part (a) shows that $(\star)$ holds without assuming differentiability. The number β used in the proof generalizes the concept of gradient (derivative) $f'(\mu)$. Any β satisfying $(\star)$ is called a **subgradient** of f at μ . For example, any $\beta \in [-1, 1]$ is a subgradient of $f(x) = |x|$ at 0.

Proof. (a). Note

$$\mu = \frac{\mu - s}{u - s}u + \frac{u - \mu}{u - s}s.$$

By convexity of f ,

$$f(\mu) \leq \frac{\mu - s}{u - s}f(u) + \frac{u - \mu}{u - s}f(s).$$

Rearranging,

$$\frac{f(\mu) - f(s)}{\mu - s} \leq \frac{f(u) - f(\mu)}{u - \mu}.$$

(b). By part (a),

$$\beta = \sup_{a < s < \mu} \frac{f(\mu) - f(s)}{\mu - s} \leq \frac{f(u) - f(\mu)}{u - \mu} < +\infty.$$

Thus

$$\frac{f(\mu) - f(s)}{\mu - s} \leq \beta \leq \frac{f(u) - f(\mu)}{u - \mu}$$

Rearranging,

$$f(s) \geq f(\mu) + \beta(s - \mu), \quad f(u) \geq f(\mu) + \beta(u - \mu)$$

Therefore,

$$f(x) \geq f(\mu) + \beta(x - \mu), \quad \forall x \in (a, b)$$

(c). Since $X \in (a, b)$, by part (b)

$$f(X) \geq f(\mu) + \beta(X - \mu).$$

Taking expectation, we obtain

$$\mathbb{E}f(X) \geq \mathbb{E}[f(\mu) + \beta(X - \mu)] = f(\mu) + \beta(\mathbb{E}X - \mu) = f(\mu) = f(\mathbb{E}X)$$

since $\mu = \mathbb{E}X$.

□

3. Is the following set convex? Show your argument.

$$S = \{\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2 : \max\{\|\mathbf{A}\mathbf{x} + \mathbf{b}\|^3, \log(1 + e^{3x_1+2x_2})\} \leq 2\}.$$

You can use any results we have proved in class.

Solution. Let

$$h_1(x) = \begin{cases} x^3, & x \geq 0 \\ 0, & x \leq 0 \end{cases}$$

Since

$$h'_1(x) = \begin{cases} 3x^2, & x \geq 0 \\ 0, & x \leq 0 \end{cases}$$

is increasing and nonnegative, h_1 is convex and increasing. Since $g_1(\mathbf{x}) = \|\mathbf{A}\mathbf{x} + \mathbf{b}\|$ is convex, by the composition rule $f_1(\mathbf{x}) = \|\mathbf{A}\mathbf{x} + \mathbf{b}\|^3$ is convex. Alternatively, we only need the fact that $x \mapsto x^3$ is convex and increasing for $x \geq 0$, since g_1 is always nonnegative.

$f_2(\mathbf{x}) = \log(1 + e^{3x_1+2x_2})$ is convex, being the composition of the convex function $h_2(x) = \log(1 + e^x)$ and the affine function $g_2(\mathbf{x}) = 3x_1 + 2x_2$. Being the maximum of two convex functions, $f(\mathbf{x}) = \max\{\|\mathbf{A}\mathbf{x} + \mathbf{b}\|^3, \log(1 + e^{3x_1+2x_2})\}$ is convex. The set S is convex since it is the 2-sublevel set of f . □

4. Determine if the following optimization problems are convex optimization problems.

(a).

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1^2 - 2x_1x_2 + x_2^2 + x_1 + x_2 \\ \text{s.t.} \quad & (x_1 - x_2)^2 + 4x_1x_2 + e^{x_1+x_2} \leq 0 \\ & x_1 - 3x_2 = 0 \end{aligned}$$

(b).

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1^2 + x_2^4 \\ \text{s.t.} \quad & x_1 e^{-(x_1+x_2)} \leq 0 \\ & x_1^2 - 2x_1x_2 + x_2^2 + x_1 + x_2 \leq 0 \end{aligned}$$

$$6x_1^2 - 7x_2 = 0$$

Solution. (a). The problem is convex. The objective function $f(\mathbf{x}) = x_1^2 - 2x_1x_2 + x_2^2 + x_1 + x_2$ is convex, since

$$\nabla^2 f(\mathbf{x}) = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix} \succeq \mathbf{O}.$$

The inequality constraint function $g(\mathbf{x}) = (x_1 - x_2)^2 + 4x_1x_2 + e^{x_1+x_2}$ is convex, since

$$\nabla^2 g(\mathbf{x}) = \begin{pmatrix} 2 + e^{x_1+x_2} & 2 + e^{x_1+x_2} \\ 2 + e^{x_1+x_2} & 2 + e^{x_1+x_2} \end{pmatrix} = (2 + e^{x_1+x_2}) \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \succeq \mathbf{O}$$

The equality constraint function $h(\mathbf{x}) = x_1 - 3x_2$ is affine.

(b). The problem is not convex. The equality constraint function $h(\mathbf{x}) = 6x_1^2 - 7x_2$ is not affine.

□